

# **POS Management and Inventory System of Nicolle Grocery Store**

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# **I. Introduction**

## **Background of the Project**

The small to medium enterprises such as Nicolle Grocery Store, maintaining efficiency and accuracy in daily operations is critical to sustaining profitability and customer satisfaction. Traditionally, the store relies on manual processes, including handwritten receipts and physical inventory counting, to manage sales and stock. While these methods may suffice for small-scale operations, they become increasingly inefficient and error-prone as the number of products, transactions, and customers grows. Human errors in recording sales, miscalculations in inventory, and delays in updating stock levels can lead to discrepancies, financial losses, and poor decision-making. Furthermore, the lack of a centralized system limits the store's ability to monitor performance and respond quickly to operational challenges. These issues highlight the need for a more reliable, automated solution that can support the store's expansion and improve overall operational control.

## **Purpose of the System**

The primary purpose of this system is to digitalize and streamline the entire sales and inventory lifecycle of Nicolle Grocery Store through an integrated Point of Sale (POS) and inventory management platform. The system is designed to provide a centralized database where all transactions, product details, and stock levels are recorded and updated in real time. It enables cashiers to efficiently process customer purchases, automatically calculate totals, and generate digital receipts, thereby reducing transaction time and minimizing errors. Additionally, the system allows administrators to manage product information, monitor inventory levels, and control user access based on roles. By integrating these core functions into a single platform, the system enhances operational efficiency, improves accuracy, and supports a more organized and scalable business process.

## **Importance of the Project**

This project plays a vital role in enabling Nicolle Grocery Store to transition toward a data-driven and technology-oriented business model. By capturing and storing detailed records of every transaction and inventory movement, the system provides valuable insights into sales trends, product performance, and stock availability. These insights empower the store owner to make informed decisions regarding pricing strategies, stock replenishment, and product selection. Moreover, the system enhances transparency and accountability by maintaining accurate and traceable records, reducing the risk of fraud and operational errors. In a competitive retail environment, adopting such a system not only improves internal efficiency but also enhances customer experience through faster and more reliable service. Ultimately, the project contributes to the long-term sustainability, growth, and competitiveness of the business.

## **Problems the system aims to solve**

- **Inefficient Checkout:** Slow manual typing is replaced by high-speed camera barcode scanning.
- **Expired Stock Waste:** Automated monitoring identifies perishable items before they expire.
- **Data Inaccuracy:** SQL Transactions ensure that sales and inventory levels stay perfectly synchronized.
- **Security Risks:** Prevents unauthorized access to sensitive financial data through role-based restrictions.

## **II. Objectives of the System**

### **General Objective**

To design and develop a cloud-synchronized Point of Sale (POS) and Inventory Management System for Nicolle Grocery Store using the Laravel framework. The system aims to automate sales processing, inventory tracking, product management, and user access control within a centralized platform. It will improve transaction efficiency, ensure accurate records, enhance data security, and generate real-time reports to support better decision-making and overall store productivity. And this aims to:

- To implement a browser-based barcode scanner using hardware-integrated camera access.

- To create an automated discount and promotion engine for marketing campaigns.
- To develop a real-time reporting suite for live monitoring of revenue and stock status.
- To implement "Soft Delete" logic to preserve historical sales data for auditing.
- To provide printable thermal-style receipts for customer transparency.

### **III. Scope and Limitations**

The system scope covers a fully functional Point of Sale (POS) terminal capable of processing sales transactions with multiple payment options, including cash and GCash, automatically calculating change, and generating printable receipts. It also includes a comprehensive inventory management module that supports full CRUD operations for product management, with the ability to upload product images either through local file storage or external image URLs. In addition, the system features a promotions module that allows the creation and execution of time-bound discount campaigns for selected products, as well as a reporting module that provides real-time sales visualization through charts and automated alerts for low-stock or threshold-based inventory monitoring.

However, the system has certain limitations. Since it is cloud-synchronized using an external MySQL service (Aiven MySQL), it requires a stable internet connection to ensure proper data storage and synchronization. The system is also dependent on compatible hardware, requiring a device with a modern web browser and a functional camera for barcode or product scanning functionality. Furthermore, digital payment transactions are not directly processed within the system; instead, payments such as GCash are recorded manually after customers complete the transaction using the GCash QR code.

## IV. System Description

### Overview

The system is a fully integrated retail management solution designed using a monolithic architecture to ensure high reliability, simplified maintenance, and consistent performance across all modules. It centralizes core retail operations into a single cohesive platform, allowing seamless interaction between sales processing, inventory management, reporting, and administrative functions without relying on distributed services. The system is developed with a modern indigo-themed user interface that emphasizes clarity, usability, and efficiency, making it suitable for both desktop and tablet environments commonly used in retail settings. Its responsive design ensures smooth navigation, fast transaction handling, and an intuitive user experience for both administrative staff and cashiers.

### Major Modules

- **Operator Terminal:** The front-facing checkout interface for cashiers.
- **Warehouse Control:** The administrative module for stock levels, pricing, and expiration tracking.
- **Marketing Engine:** A module for managing discounts and promotional campaigns.
- **Analytics Suite:** An executive dashboard for financial overview.

### User Roles

- **Administrator:** Full access to all modules, including staff management, financial reports, and system settings.
- **Cashier:** Access is strictly limited to the Terminal module for processing sales.

## V. Technologies Used

- **Laravel 12:** The primary backend framework used for its robust security, Eloquent ORM, and routing capabilities.
- **MySQL (Aiven Cloud):** The relational database management system used for secure, remote data storage.
- **Tailwind CSS:** A utility-first CSS framework used to create the professional "Enterprise" design.

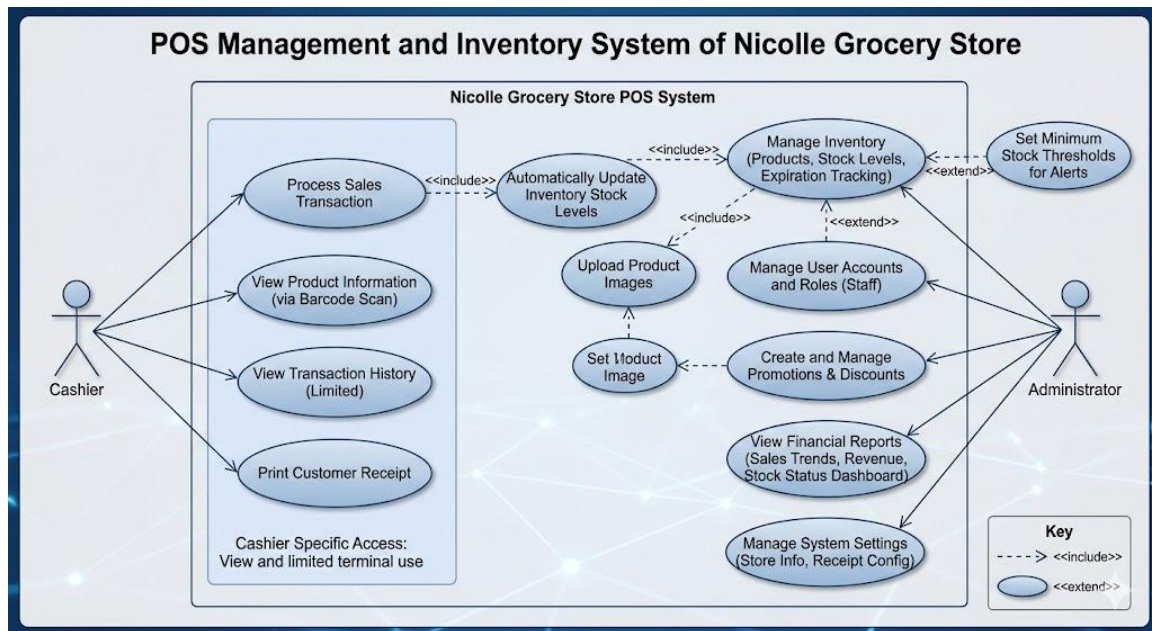
- Alpine.js: A lightweight JavaScript framework used to handle real-time terminal logic (cart, clock, and scanner).
- HTML5-QRCode: An advanced library utilized to turn standard device cameras into professional barcode scanners.
- Chart.js: Used for rendering dynamic, real-time sales trend graphs.

## **VI. System Design**

### **Use Case Diagram**

The POS Management and Inventory System of Nicolle Grocery Store is designed to streamline and integrate the store's sales and inventory operations into a single efficient platform. The system has two main users: the Cashier and the Administrator. The Cashier is responsible for handling daily point-of-sale activities, which include processing sales transactions, viewing product information through barcode scanning, checking limited transaction history, and printing customer receipts. When a sales transaction is processed, the system automatically updates the inventory stock levels to ensure real-time accuracy of product availability.

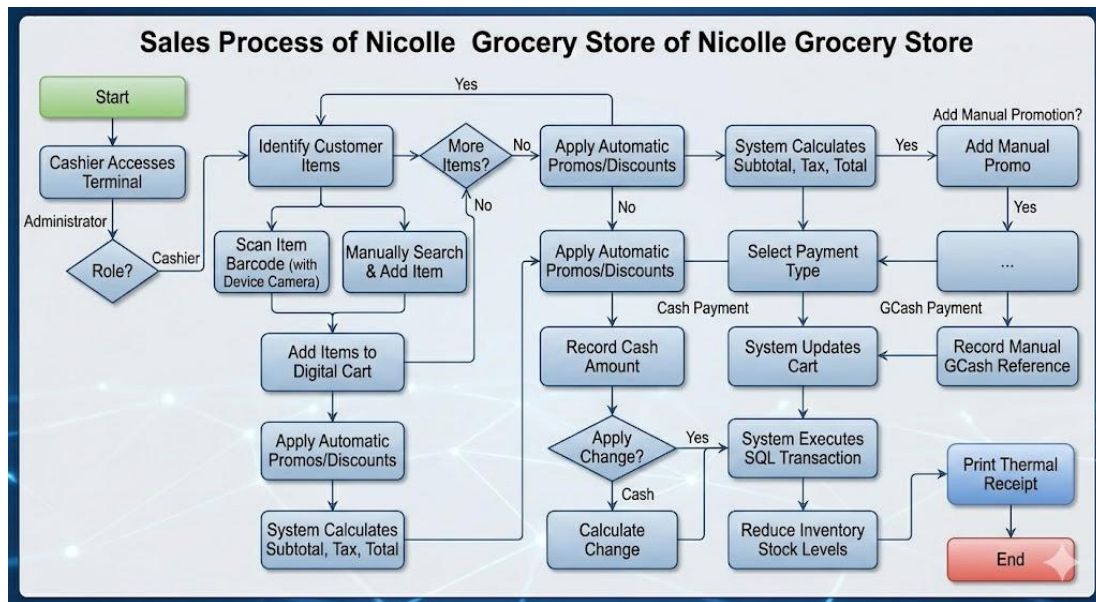
On the other hand, the Administrator has full access to the system's advanced features and overall management functions. The Administrator manages the inventory by adding, updating, and tracking products, including stock levels and expiration dates. They can also set minimum stock thresholds to trigger alerts when items are running low. In addition, the Administrator is responsible for managing user accounts and roles, ensuring proper access control for staff members. The system also allows the creation and management of promotions and discounts, uploading and assigning product images, and monitoring financial reports such as sales trends, revenue, and stock status through a dashboard. Furthermore, the Administrator can configure system settings, including store information and receipt formatting. Overall, the system ensures efficient operations by automating inventory updates during sales and providing structured access depending on user roles.



**Figure 1: Use Case Diagram**

## Flowchart Diagram

The sales process at Nicolle Grocery Store begins with the cashier logging into the terminal, where the system verifies their Role to grant access. Once authenticated, the cashier enters an iterative loop to Identify Customer Items by either scanning barcodes or performing manual searches, adding each to a digital cart while the system automatically calculates subtotals, taxes, and promotions. After all items are processed and any manual discounts are applied, the cashier selects a Payment Type. For Cash transactions, the system calculates the necessary change, while for GCash, it prompts for a manual reference number. Upon payment confirmation, the system executes a formal SQL Transaction to ensure data integrity, simultaneously updating the database to Reduce Inventory Stock Levels. The process concludes with the generation of a Thermal Receipt and the termination of the transaction session.

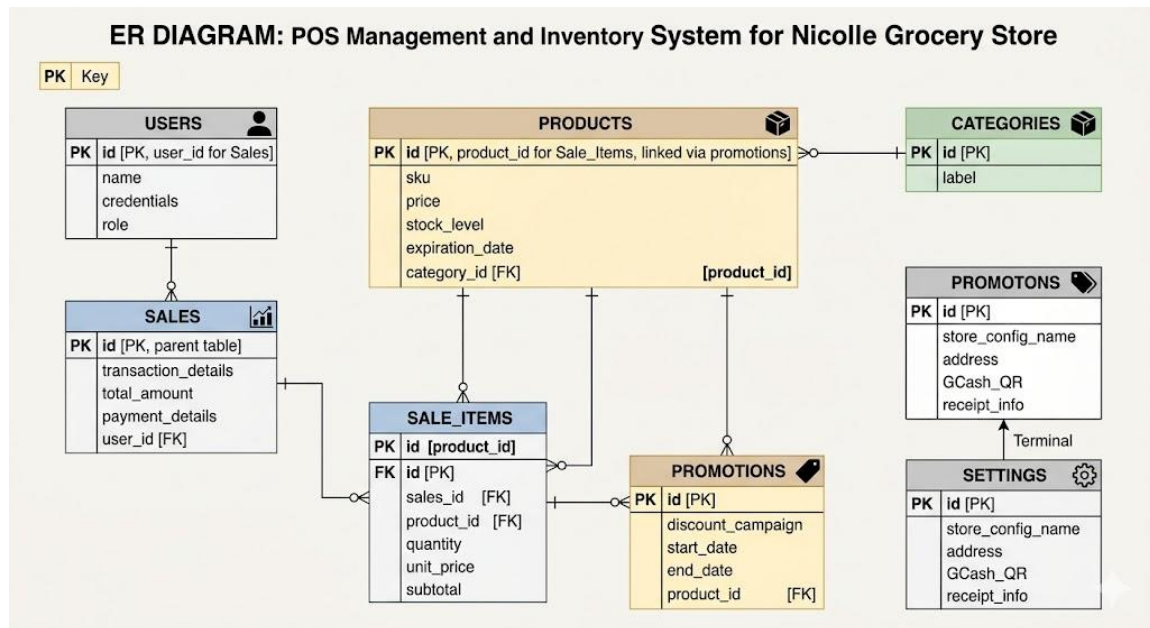


**Figure 2: Flowchart Diagram**

## ER Diagram

The database structure for Nicolle Grocery Store integrates five core modules into a cohesive, cloud-synchronized relational model. Staff accountability is enforced by a One-to-Many relationship between **USERS** and **SALES**, tracing every transaction to a specific cashier. Inventory is classified via a One-to-Many link between **CATEGORIES** and **PRODUCTS**, streamlining warehouse control. The core transaction engine utilizes **SALE\_ITEMS** as a bridge table to resolve a Many-to-Many relationship between **SALES** and **PRODUCTS**; this captures real-time receipt line items and automatically triggers stock deductions upon checkout. For the marketing engine, the **PROMOTIONS** table connects to **PRODUCTS** in a One-to-Many setup, allowing the terminal to automatically apply active, time-bound discounts during a scan. Finally, the independent **SETTINGS** table stores global configurations—such as the store address and GCash QR code—and is accessed directly by the terminal to generate printable thermal receipts without the need for complex foreign key relationships.

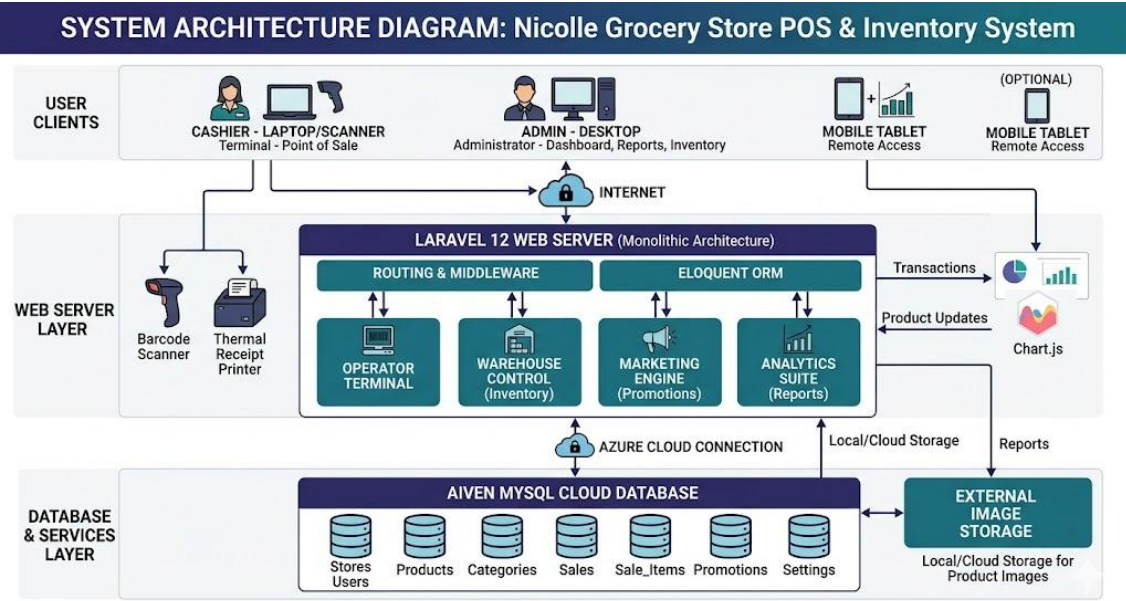




**Figure 3: Er Diagram**

## System Architecture Diagram

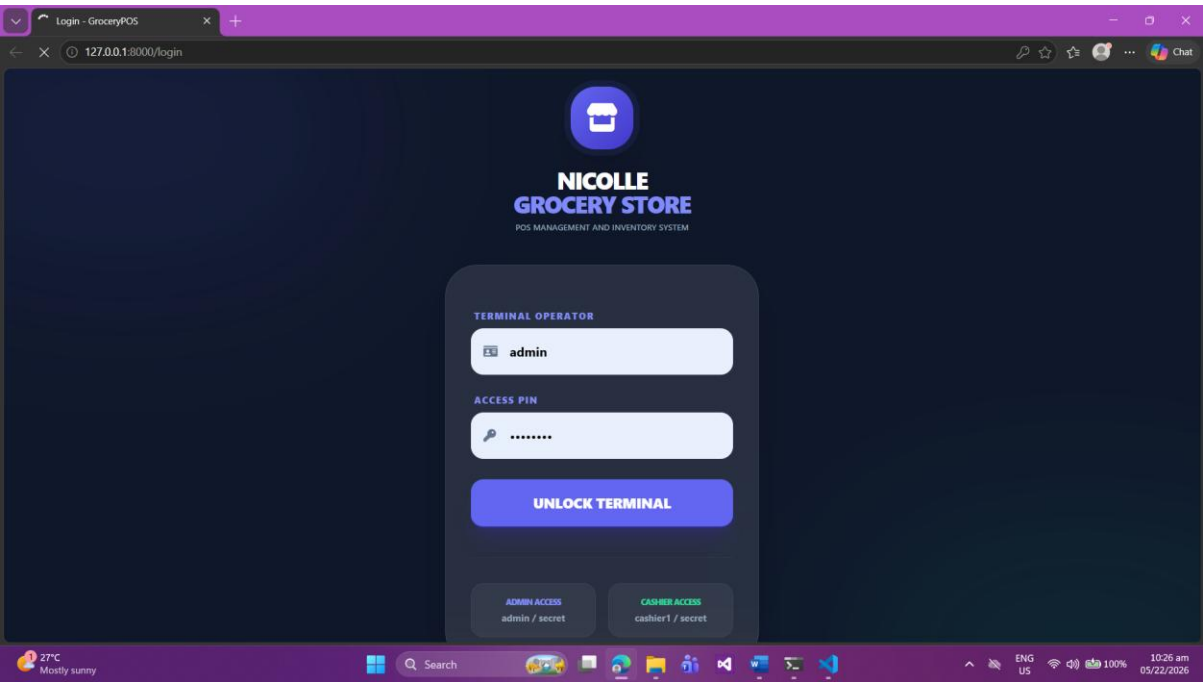
The system architecture for the Nicolle Grocery Store Point of Sale and Inventory System follows a structured, three-tier monolithic design based on the Model-View-Controller (MVC) pattern. At the User Clients Layer, cashiers and administrators access the application via desktop, laptop, or mobile tablet browsers styled with Tailwind CSS. The cashier's environment directly coordinates with physical hardware, leveraging the HTML5-QRCode library to transform device cameras into barcode scanners alongside thermal receipt printers for customer transparency. These clients connect over the internet to the Web Server Layer, driven entirely by the Laravel 12 framework. Within this single server application, built-in routing and middleware enforce role-based access restrictions—such as preventing cashiers from accessing administrative financial data—while an Eloquent ORM coordinates server logic across the four core modules: the Alpine.js-powered Operator Terminal, Warehouse Control, the Marketing Engine, and the Analytics Suite, which passes dynamic data to Chart.js for real-time sales rendering. Finally, at the Database and Services Layer, the application communicates through a secure cloud connection to an external Aiven MySQL service. This cloud database houses the system's seven structural tables (users, products, categories, sales, sale\_items, promotions, and settings), keeping historical transaction items safely preserved via "Soft Delete" logic while referencing external or local image storage for product listings.



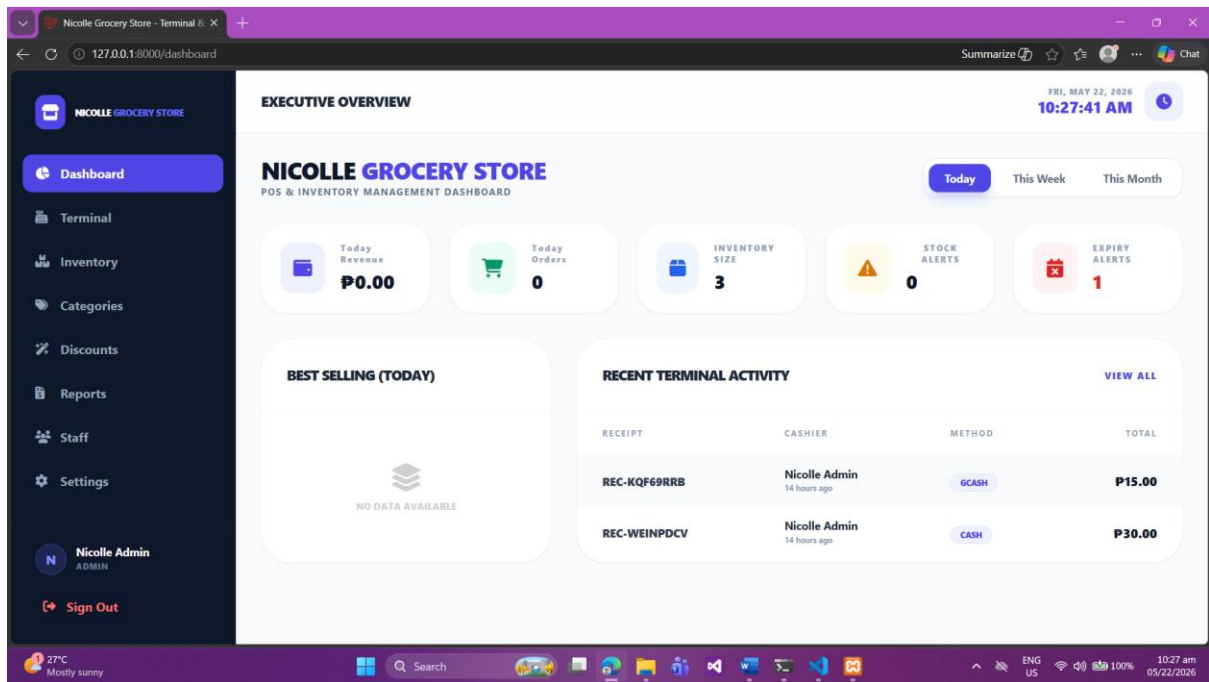
**Figure 4:** System Architecture Diagram

**Interface Design/Screenshot**

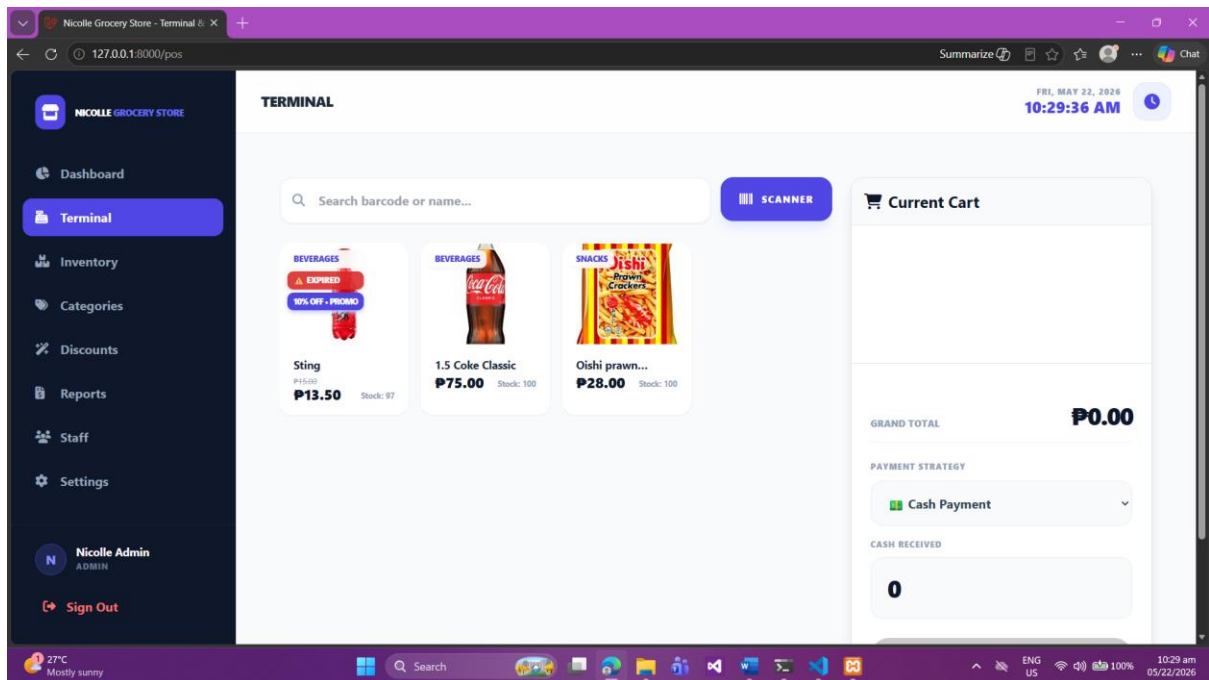
**(Admin Account)**



**Figure 5:** Log in Page



**Figure 6: Admin Dashboard**



**Figure 7: Admin Terminal**

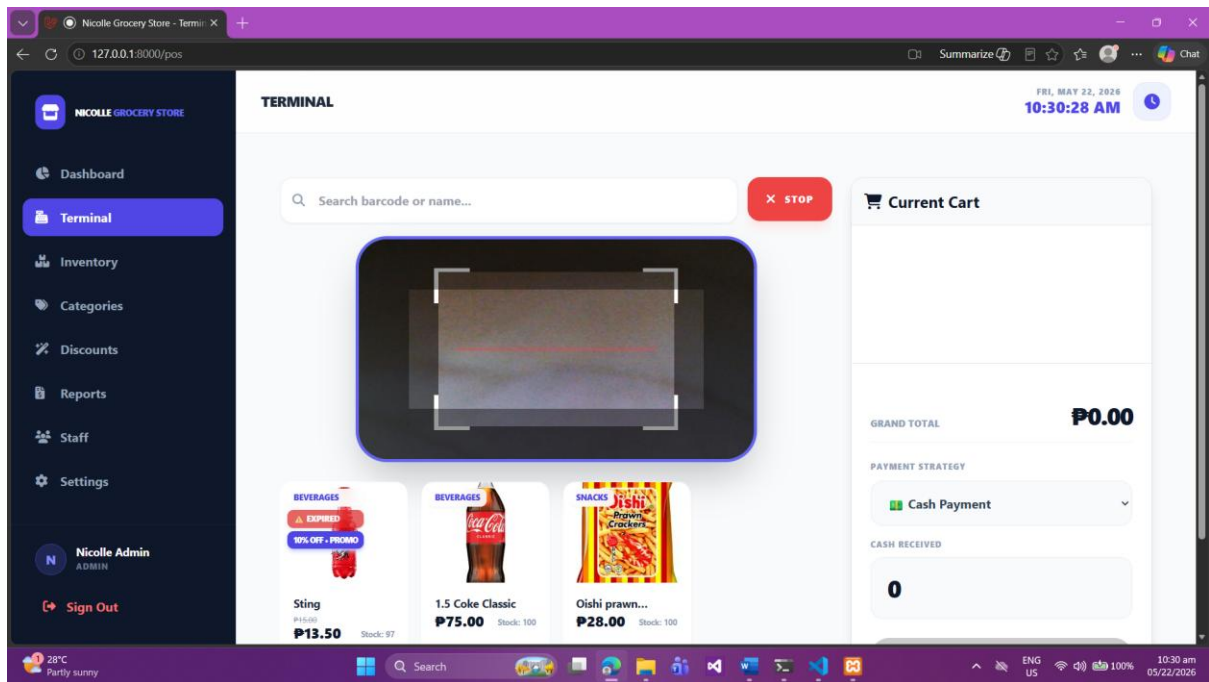


Figure 8: Admin Terminal/ Bar Code Scanner

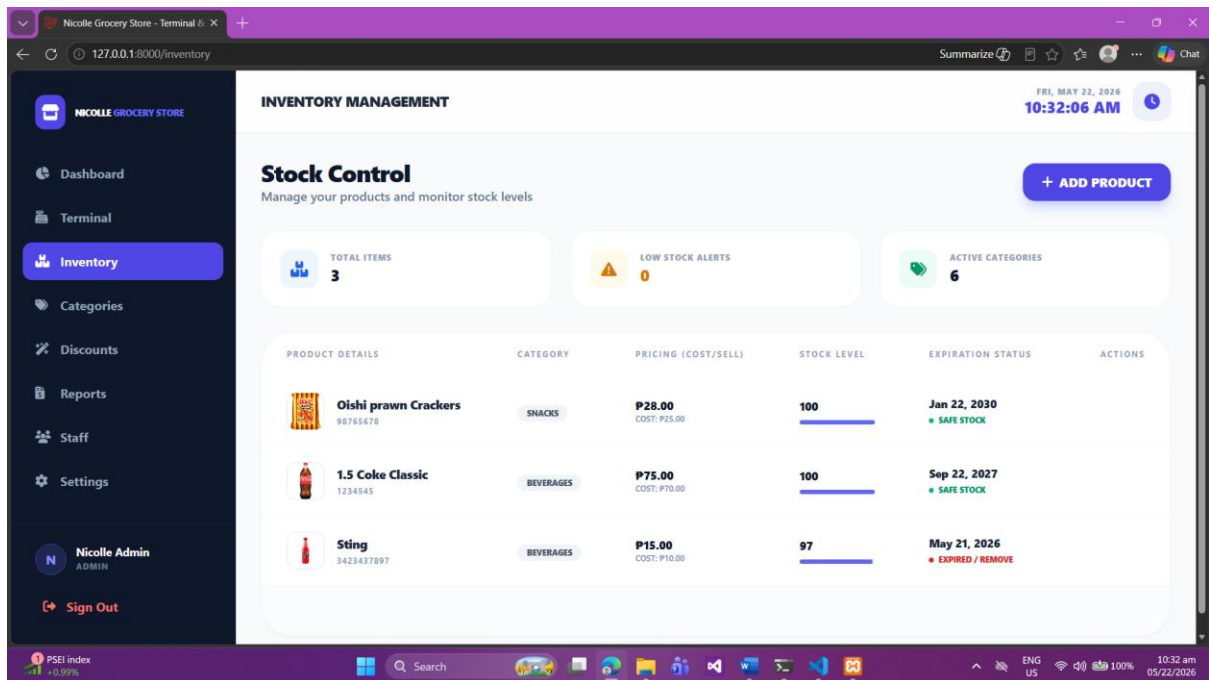


Figure 9: Admin Inventory

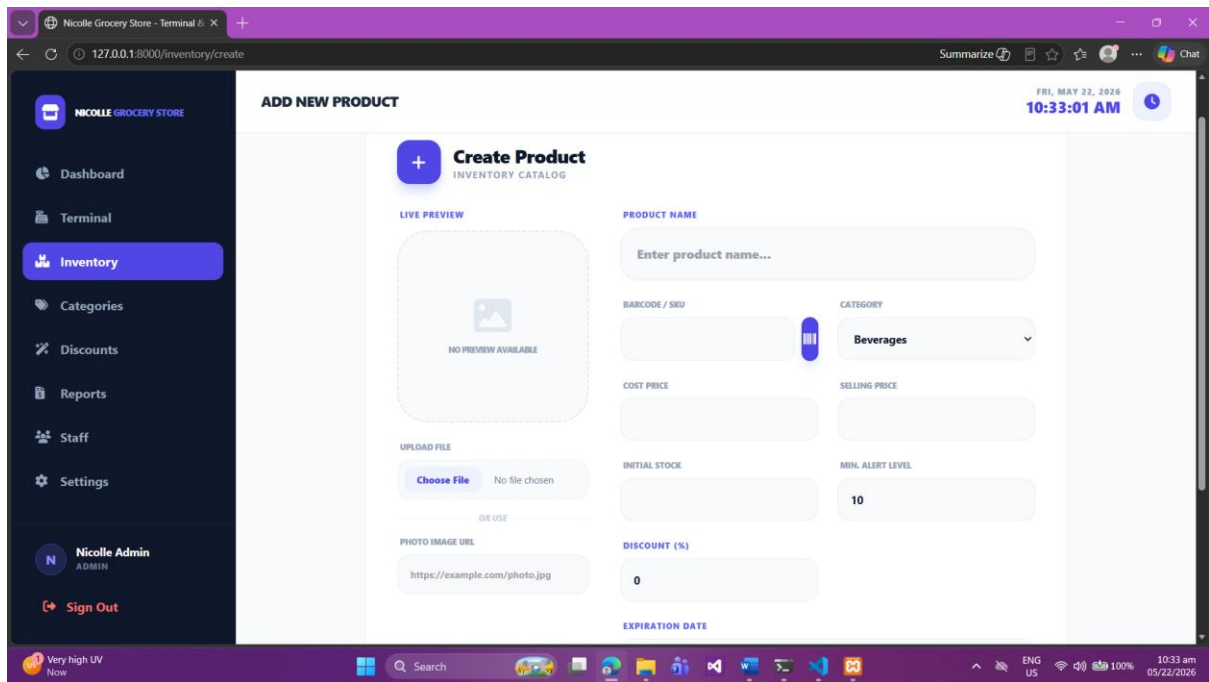


Figure 10: Admin Inventory / Add new Product

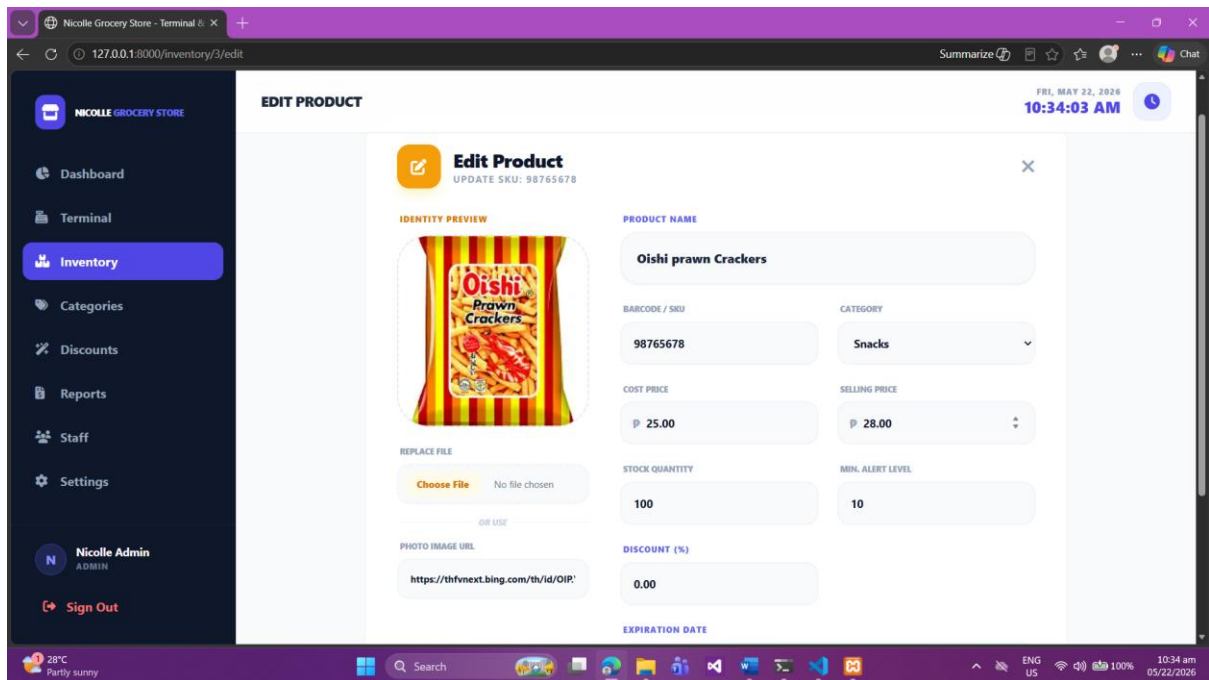


Figure 11: Admin Inventory / Edit Product

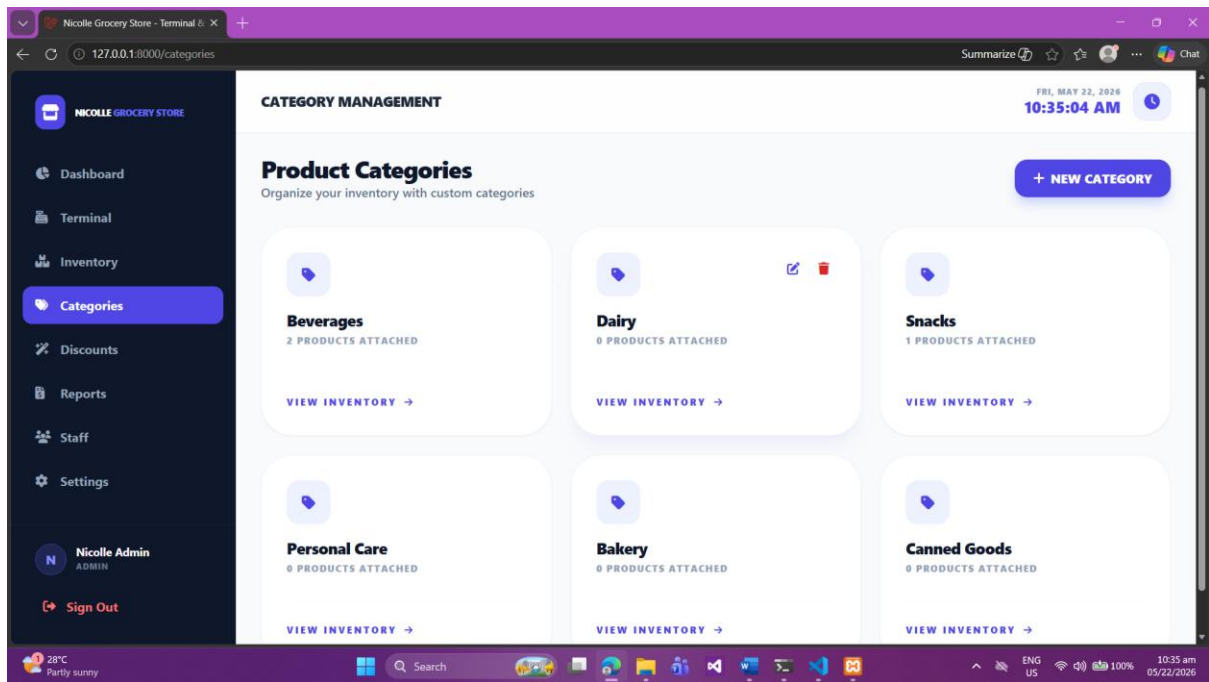


Figure 12: Admin Categories

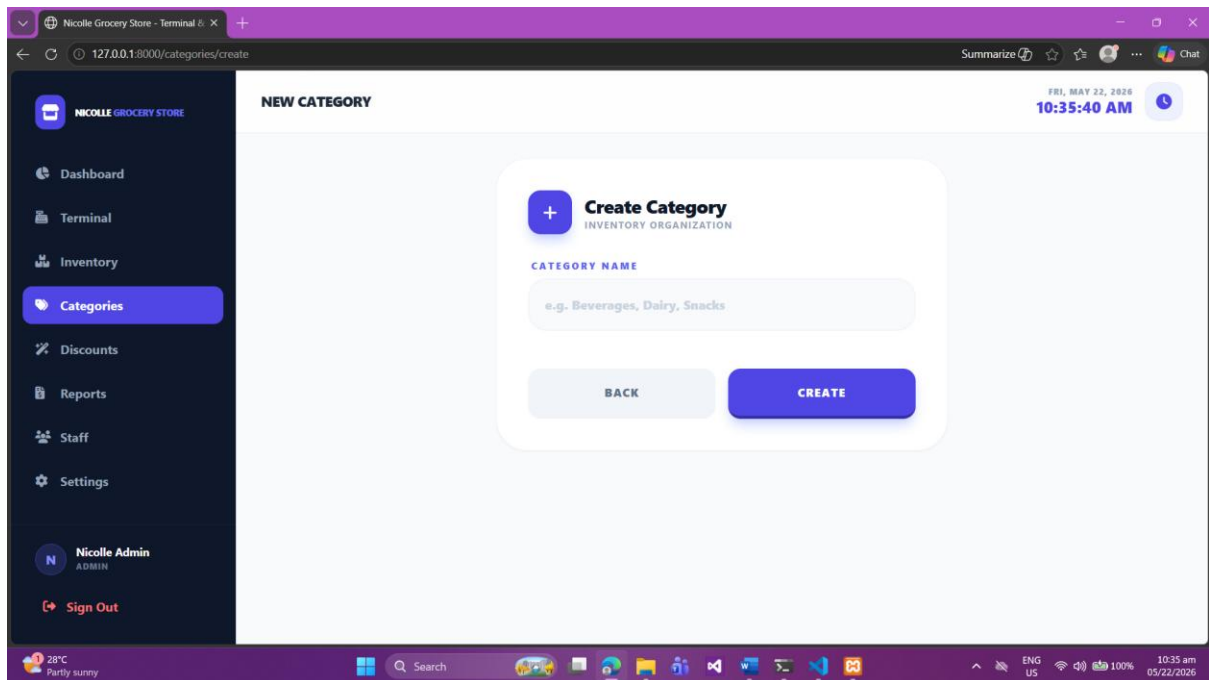


Figure 13: Admin Categories / Create new Category



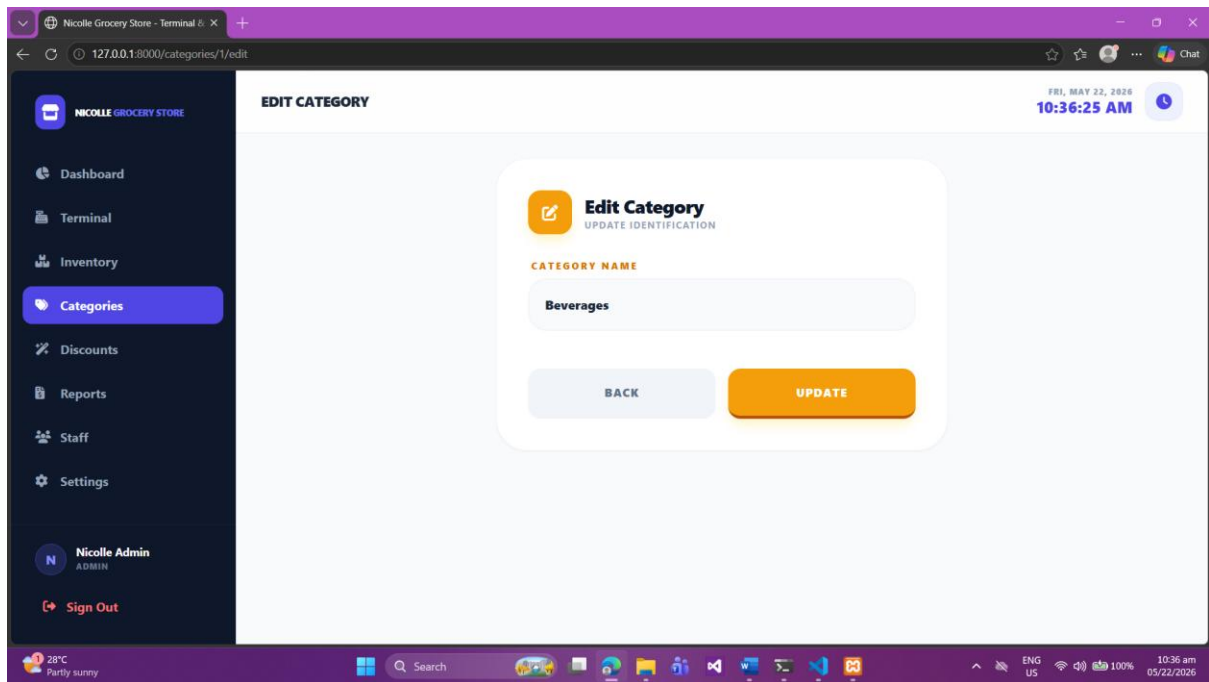


Figure 14: Admin Categories / Edit Category

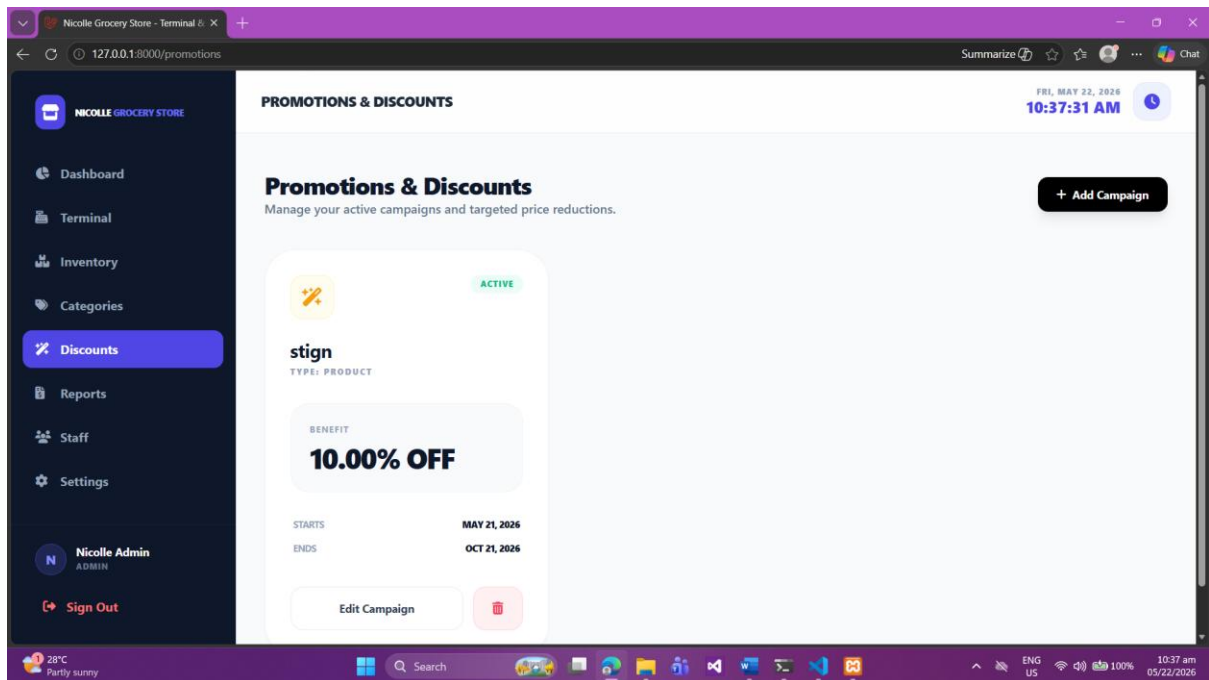


Figure 15: Admin Discounts

The screenshot shows the 'NEW PROMOTION CAMPAIGN' form in the Nicole Grocery Store admin interface. The form is titled 'NEW CAMPAIGN' with the subtitle 'PROMOTION STRATEGY'. It includes the following fields and controls:

- CAMPAIGN NAME:** A text input field with the placeholder 'e.g. summer sale, weekend blast'.
- APPLY TO PRODUCT:** A dropdown menu with the placeholder 'Select Target Item'.
- DISCOUNT PERCENTAGE:** A text input field with the value '10'.
- START DATE:** A date picker showing '05/22/2026'.
- END DATE:** A date picker showing '06/22/2026'.
- Buttons:** 'CANCEL' and 'LAUNCH CAMPAIGN'.

The interface includes a sidebar with navigation links: Dashboard, Terminal, Inventory, Categories, Discounts (highlighted), Reports, Staff, and Settings. The user is logged in as 'Nicolle Admin'.

Figure 16: Admin Discounts / Add new Campaign

The screenshot shows the 'EDIT CAMPAIGN' form in the Nicole Grocery Store admin interface. The form is titled 'Edit Campaign' with the subtitle 'NICOLLE GROCERY PROMOTION ENGINE'. It includes the following fields and controls:

- CAMPAIGN NAME:** A text input field with the value 'stign'.
- APPLY TO PRODUCT:** A dropdown menu with the value 'Sting'.
- DISCOUNT (%):** A text input field with the value '10.00'.
- START DATE:** A date picker showing '05/21/2026'.
- END DATE:** A date picker showing '10/21/2026'.
- Buttons:** 'BACK' and 'SAVE CHANGES'.

The interface includes a sidebar with navigation links: Dashboard, Terminal, Inventory, Categories, Discounts (highlighted), Reports, Staff, and Settings. The user is logged in as 'Nicolle Admin'.

Figure 17: Admin Discounts / Edit Campaign



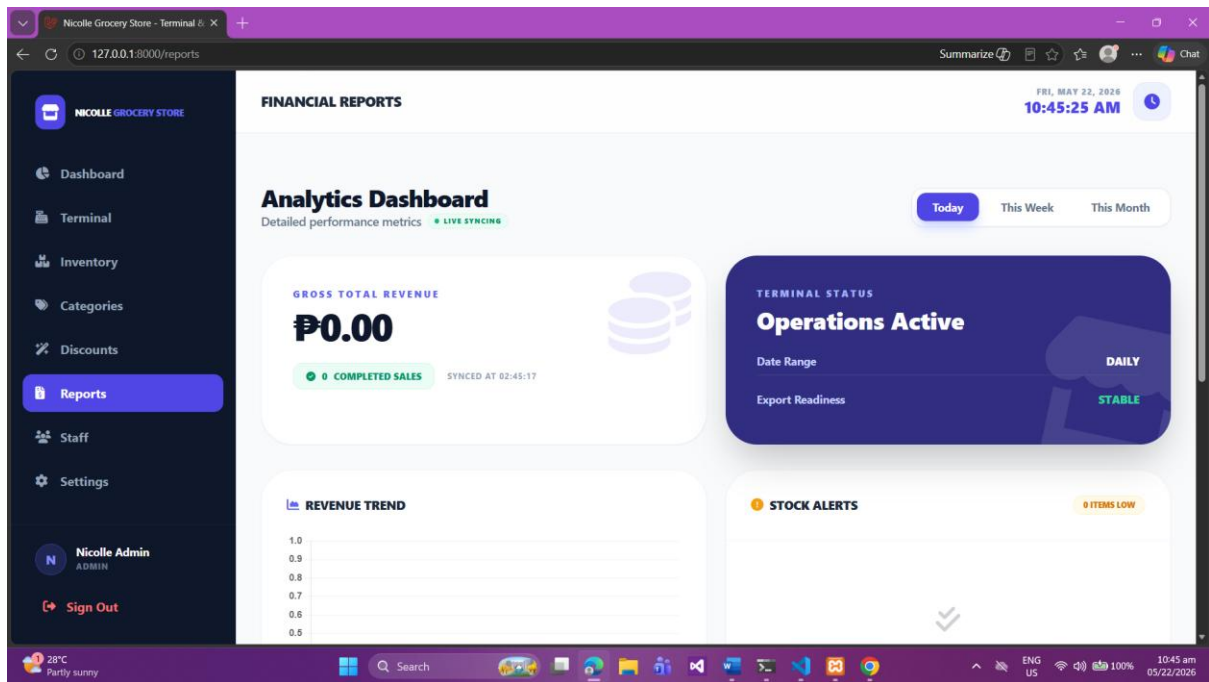


Figure 18: Admin Reports

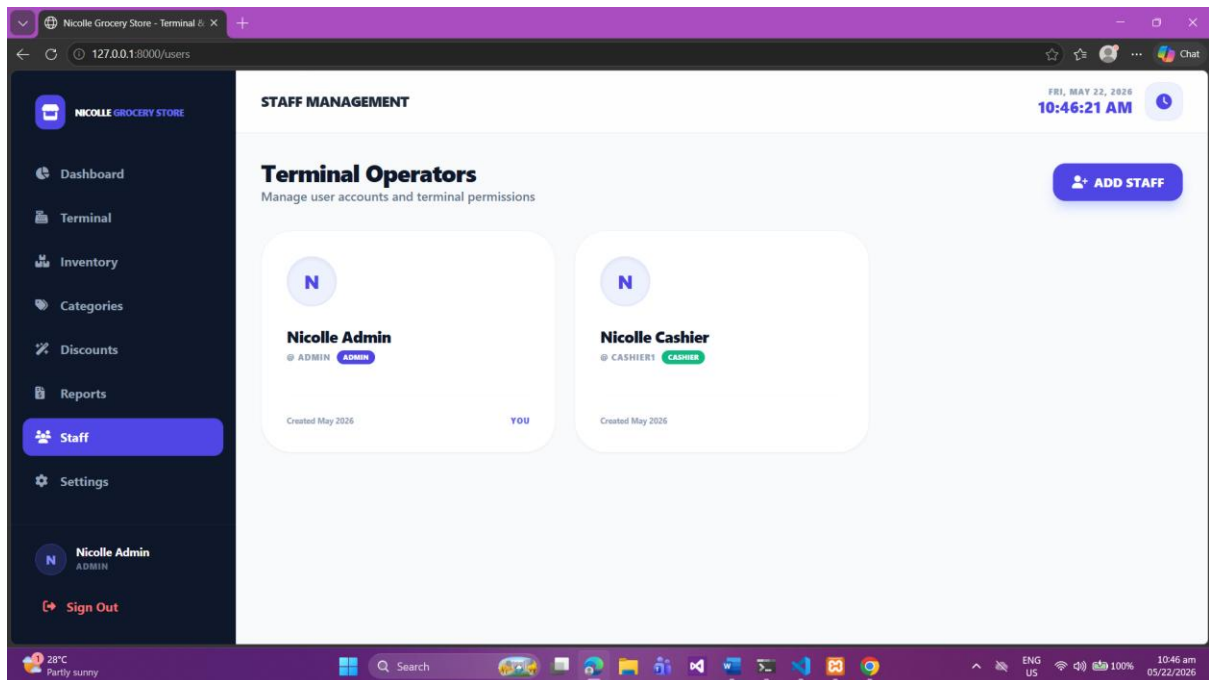
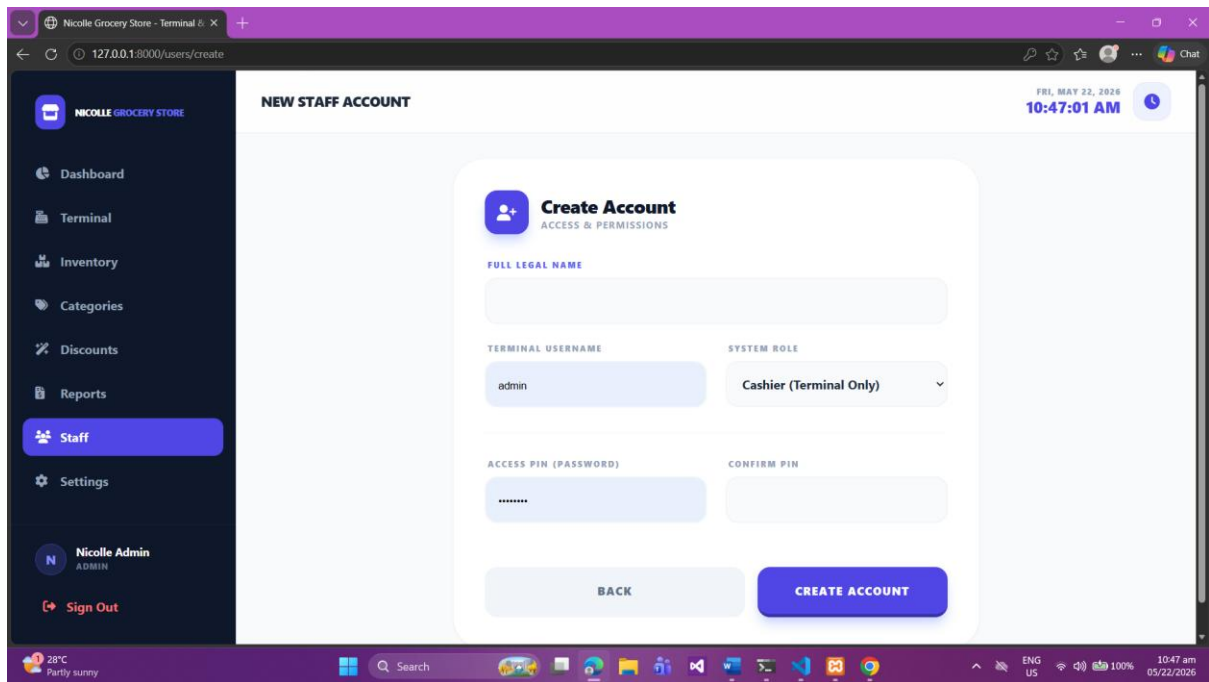
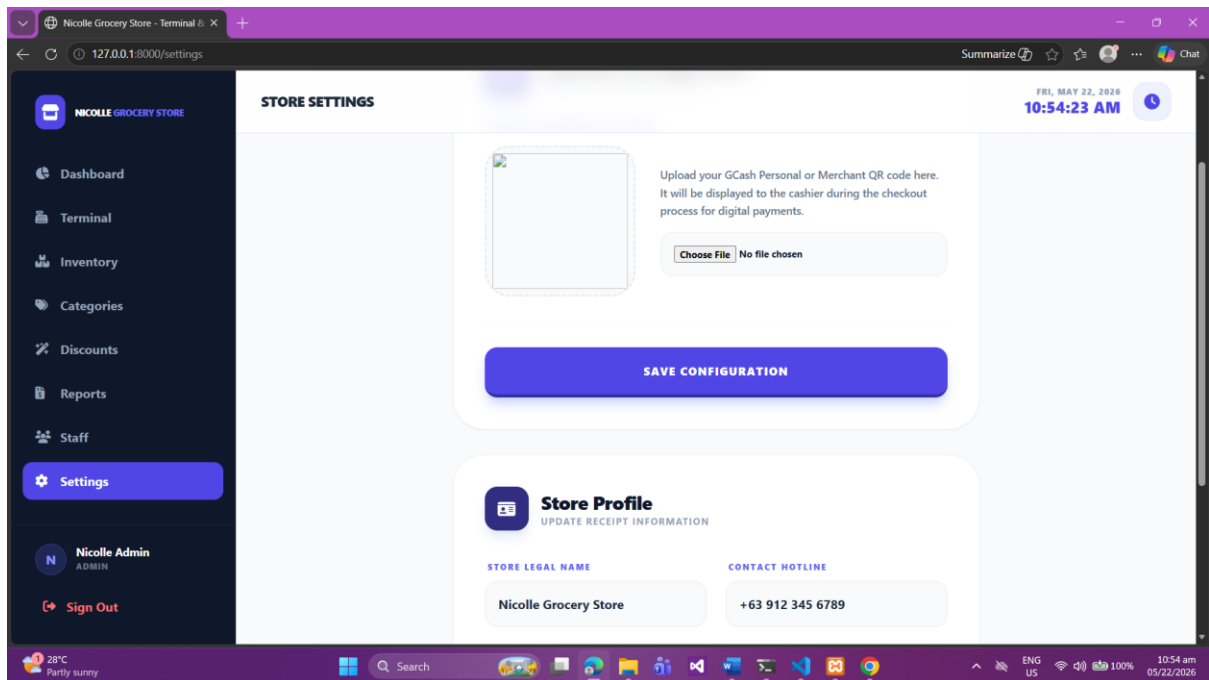


Figure 19: Admin Staff



**Figure 20: Admin Staff/ Add new Staff Account**



**Figure 21: Admin Settings**

**(Cashier Account)**

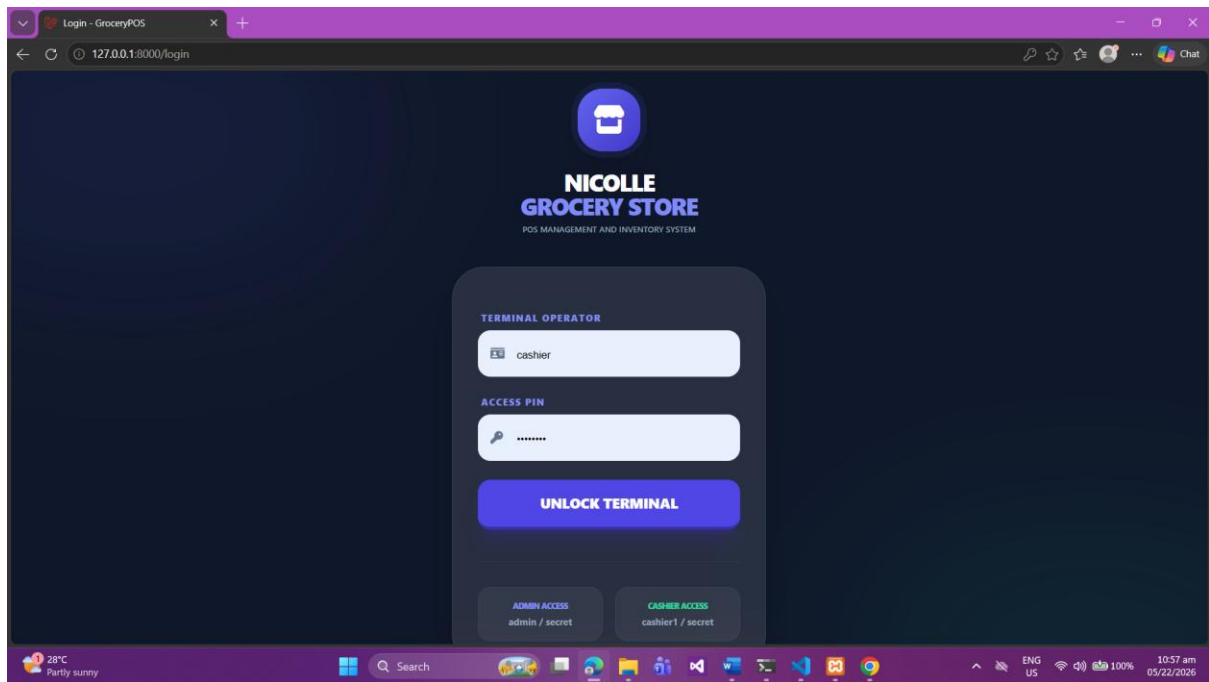


Figure 22: Cashier log in form

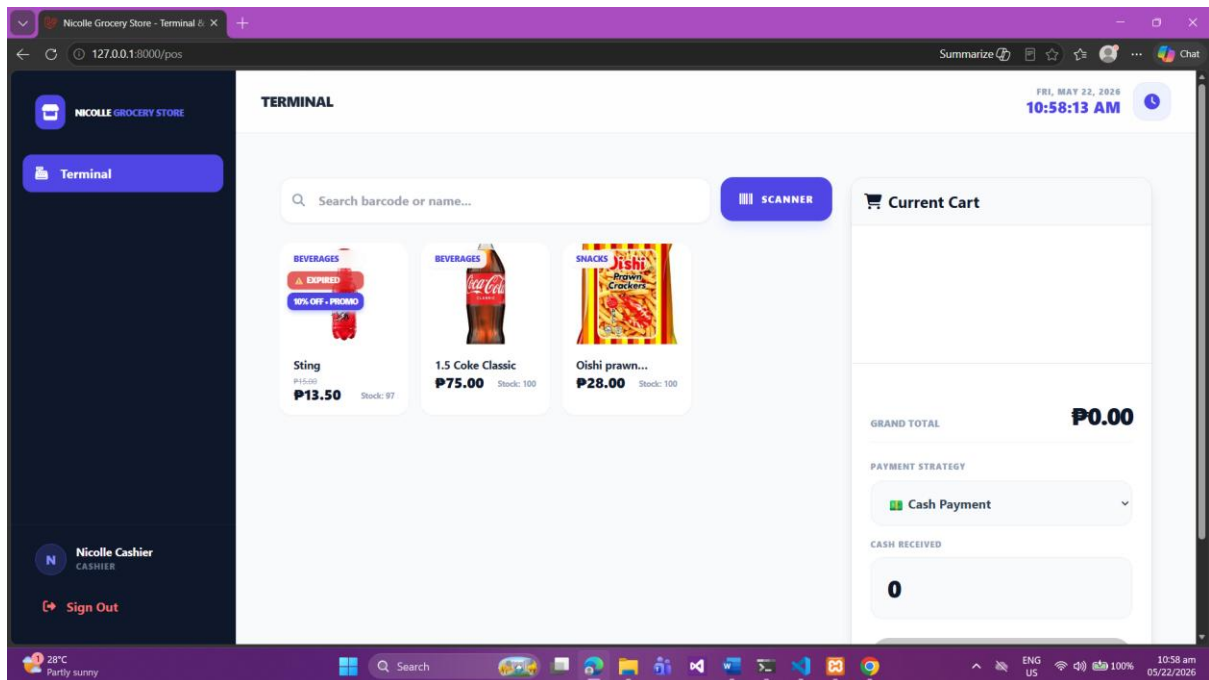
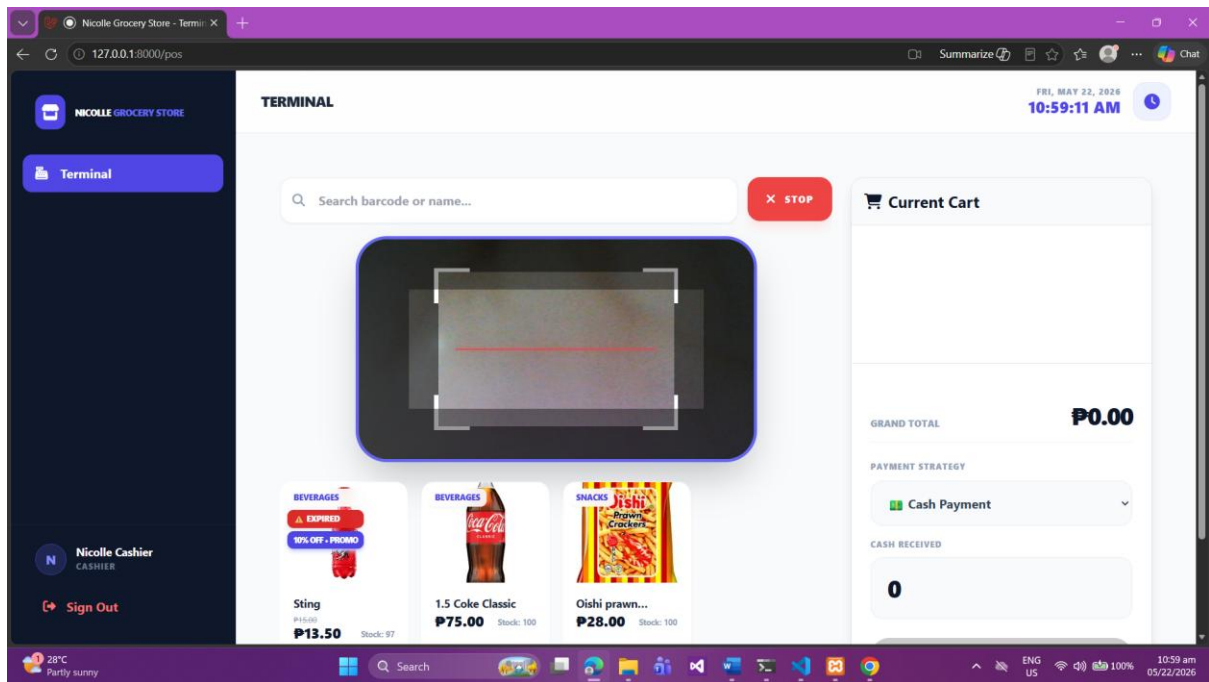


Figure 23: Cashier Terminal



**Figure 24:** *Cashier Barcode Scanner*

## VII. Database Structure

| Table             | Field Description                                                               | Relationships                                |
|-------------------|---------------------------------------------------------------------------------|----------------------------------------------|
| <b>users</b>      | Stores staff credentials, names, and roles.                                     | Primary key for Sales (user_id).             |
| <b>products</b>   | Contains item SKUs, prices, stock levels, and expiration dates.                 | Linked to Sale Items and Categories.         |
| <b>categories</b> | Stores inventory classification labels.                                         | One-to-Many relationship with Products.      |
| <b>sales</b>      | Master record of all completed transactions.                                    | Parent table of Sale Items.                  |
| <b>sale_items</b> | Stores individual receipt line items (product, quantity, unit price, subtotal). | Links Sales and Products.                    |
| <b>promotions</b> | Manages discount campaigns with start and end dates.                            | Applied to specific Products.                |
| <b>settings</b>   | Stores global store configuration (name, address, GCash QR, receipt info).      | Accessed by Terminal for receipt generation. |

## VIII. Implementation and Testing

## **Development Process**

The system was developed using the Agile Methodology, following an iterative and incremental approach to ensure continuous improvement and adaptability throughout the development lifecycle. The process began with the design and implementation of the core database structure through migrations, establishing the foundational entities such as users, products, sales, and inventory relationships. This was followed by the development of essential backend logic and functionality, particularly the Point-of-Sale Terminal, which handles transaction processing, payment handling, and receipt generation. Subsequent iterations focused on expanding system features, including inventory management, promotions, and user role access control, while continuously refining system performance and usability. In the later stages, emphasis was placed on enhancing reporting capabilities through real-time analytics and dashboard visualization, as well as improving the overall user interface and branding to ensure a professional and consistent retail experience.

## **Testing Procedures**

- Security Testing: Verified that Cashiers are automatically redirected if attempting to access /reports.
- Integrity Testing: Performed "Double-Checkout" tests to ensure inventory count never goes negative.
- Cross-Browser Testing: Confirmed barcode scanner functionality on both Windows (VS Code Environment) and Android mobile devices.

## **Problems and Solutions**

- Problem: Deleting a product caused historical sales reports to crash.
- Solution: Implemented Soft Deletes in the Product model, allowing items to be archived while retaining their identity in past sales records.

## IX. Conclusion

The POS Management and Inventory System for Nicolle Grocery Store successfully achieved all defined project objectives by delivering a fully integrated and efficient retail management solution. The system establishes a seamless connection between inventory management and the Point of Sale terminal, ensuring that product stocks are automatically updated in real time after every transaction. Built using Laravel 12 and a cloud-based MySQL database, it provides a secure, scalable, and centralized platform that supports reliable data storage and synchronization across all operations. The system enhances operational efficiency by streamlining sales processing, improving inventory accuracy, and reducing manual errors. It also strengthens business decision-making through real-time reporting and analytics, allowing the store to monitor sales performance and stock levels effectively. Overall, the system equips Nicolle Grocery Store with a modern retail infrastructure comparable to larger commercial establishments, promoting long-term growth, improved productivity, and enhanced data security.

## X. Recommendations

- **Multi-Store Support:** Enhance the database to support multiple branches with a single administrative login.
- **SMS Receipts:** Integrate an SMS API to send digital receipts to customers' mobile numbers.
- **Automatic Re-ordering:** Implement a module that automatically generates purchase orders when stock hits the minimum threshold.
- **Customer Loyalty:** Add a point-based system to reward frequent shoppers at Nicolle Grocery Store.